

Preliminary Dataset Report

1. WHAT WAS RECEIVED

The dataset delivery consists of 4 video files, no manifest, and no ground-truth labels file. This differs from the Day 4 assumed schema (which expected a manifest.csv and a labels/ folder) — flagged as an open item for the mentor.

File	Resolution	Frame Rate	Reported Frames	Duration
1LED_92bps.mp4	640 x 360	260 fps	30,058	115.6 s
2LEDs_92bps.mp4	640 x 360	260 fps	21,157	81.4 s
3LEDs_92bps.mp4	640 x 360	260 fps	13,057	50.2 s
4LEDs_92bps.mp4	640 x 360	260 fps	11,033	42.4 s

File naming encodes the transmitter configuration: N simultaneous LEDs at a 92-bps modulation rate. This is the only metadata available, everything else below was derived directly from the video files themselves.

2. CORRUPTION / READABILITY CHECK

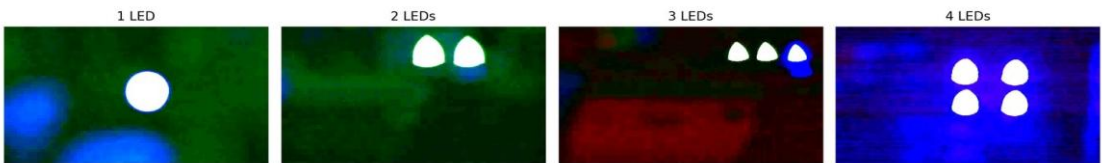
Every frame in every file was read sequentially, start to finish, with no seeking. This is the strongest possible check for corruption or truncation short of a full checksum comparison.

File	Reported Frames	Result
1LED_92bps.mp4	30,058	All 30,058 frames read cleanly — 0 mismatch
2LEDs_92bps.mp4	21,157	All 21,157 frames read cleanly — 0 mismatch
3LEDs_92bps.mp4	13,057	All 13,057 frames read cleanly — 0 mismatch
4LEDs_92bps.mp4	11,033	All 11,033 frames read cleanly — 0 mismatch

Result: no corruption in any of the 4 files. This risk from the Day 4 checklist is closed.

3. VISUAL SPOT-CHECK

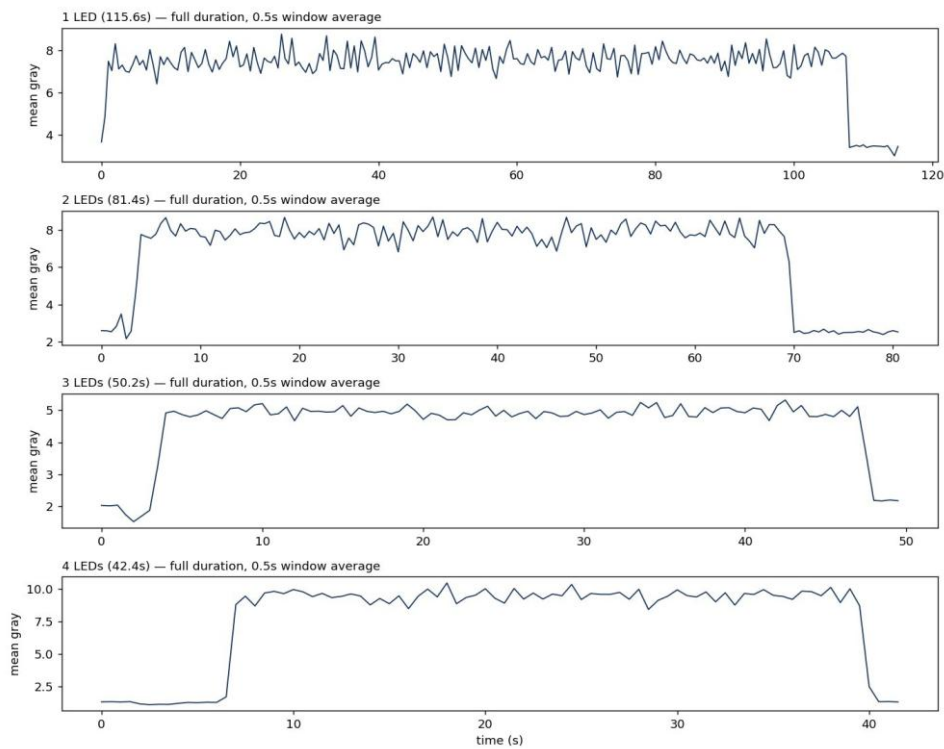
A brightness-boosted frame from each video (raw OCC frames are very dark, mean pixel value around 5–10 out of 255, since most of the frame is unlit background; this gain is a display transform only, the underlying frame data is untouched).



The transmitter LEDs are clearly visible and well-separated from the background in all four videos. Worth noting: the background colour cast differs across the four files (greenish in the 1–2 LED videos, reddish in the 3 LED video, bluish in the 4 LED video) — likely different white-balance or lighting conditions between recording sessions rather than a single controlled capture batch. This is a new item for the Day 4 risk checklist, not something originally anticipated. The 3-LED frame above also happens to show one of the three LEDs off mid-frame, early visual confirmation that OOK on/off modulation is present in the raw footage.

4. FIRST-LOOK SIGNAL: FULL-DURATION OVERVIEW

Whole-frame mean pixel intensity, in 0.5-second windows, across the entire duration of each file.



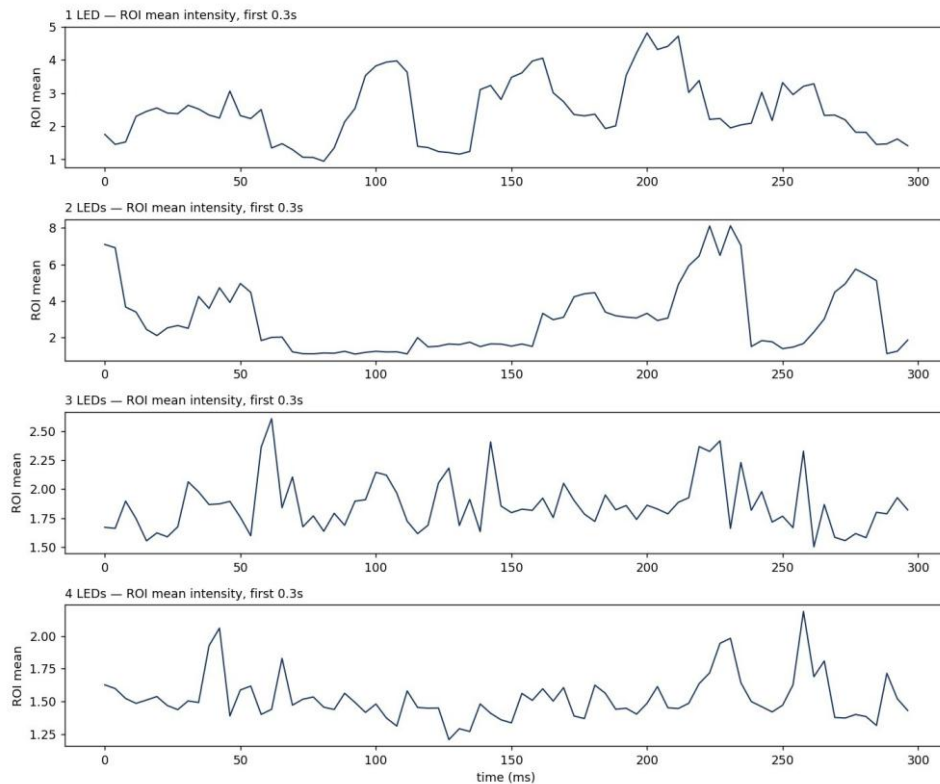
Finding not in the Day 4 assumptions: every video has a dark lead-in and lead-out period before/after the actual transmission. Quantified by thresholding against the plateau level:

File	Active Window	Lead (dead time)	Trail (dead time)
1LED_92bps.mp4	0.5s → 107.5s	0.5 s	8.1 s
2LEDs_92bps.mp4	3.5s → 69.5s	3.5 s	11.9 s
3LEDs_92bps.mp4	3.5s → 47.5s	3.5 s	2.7 s
4LEDs_92bps.mp4	7.0s → 39.5s	7.0 s	2.9 s

These lead/trail windows should be trimmed (or explicitly masked) before symbol decoding in Day 9–11 preprocessing, decoding against them would just be noise.

5. IS OOK MODULATION ACTUALLY VISIBLE?

A 300ms window from inside each active period, cropped to the region around the brightest detected blob rather than the whole frame (whole-frame mean is dominated by background and washes the signal out).



A clear modulation envelope is visible once cropped to the LED region — confirming the signal is genuinely present and usable — but it is not yet a clean bit-level 0/1 trace at this stage. Getting from here to reliable symbol decoding needs per-row (rolling-shutter) intensity extraction and explicit symbol synchronization, which is Day 9-13 work, not a problem with the raw data itself.

6. CLASS COVERAGE & DUPLICATES

One video per transmitter condition (1, 2, 3, and 4 simultaneous LEDs), 4 files total, no repeats or near-duplicates (confirmed by distinct file sizes and durations). This reframes the Day 4 "condition imbalance" risk: it isn't an imbalance across many samples, it's a small-sample-size limitation, one data point per condition. Any claim about how decoding accuracy scales with LED count will need to be stated as a single-sample observation, not a statistically robust trend, unless more data arrives later.

7. DAY 4 RISK CHECKLIST — STATUS AFTER INSPECTION

Day 4 Assumption	Status After Day 7
manifest.csv with fps/resolution/labels	Not present. All metadata pulled directly from video file headers instead.
Ground-truth bit sequence / labels	Not present in this delivery. BER scoring is not possible yet — flagged for the mentor.
Missing / corrupt files	Closed. All 4 files fully readable; frame counts match exactly.
Condition imbalance	Reframed: 1 video per condition, a small-sample limitation rather than a skew across many samples.
Temporal / fps consistency	Confirmed: 260 fps constant within and across all 4 files.
Leakage risk	Not applicable yet — no duplicate or repeated videos in this delivery.

(New — not anticipated)	Dead time before/after active transmission in every video; needs trimming logic before decoding.
(New — not anticipated)	Background colour cast differs across the 4 files, possibly inconsistent capture/lighting conditions between sessions.

8. NEXT STEP

Day 8's video/signal-quality audit builds directly on these findings: the active-window timestamps found here define where to look for symbols, and the colourcast/lead-trail observations feed into the video-quality risk table due that day.